Yugandhar Narravula

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SUMMARY

AI/LLM Developer who ships production Generative AI and Agentic AI systems in enterprise environments — AI agents, copilots, RAG assistants, and workflow automation that stay reliable when they run unattended. Two years in Python building LangGraph and LangChain systems (an 11-agent orchestration graph with typed shared state, MCP tool integrations, and graph-level error recovery, plus a deployed RAG service) on a strong software-engineering foundation: Python, FastAPI, REST APIs, SQL, and Git/GitHub. Owns the full loop from concept through the AI evaluation frameworks, deployment, and monitoring that decide whether an agent workflow actually works.

CORE SKILLS

Agent Engineering: LangGraph, LangChain, Production AI Agents, Multi-Agent Systems, Copilots & Conversational Assistants, Workflow Automation, Tool Orchestration & Function Calling, Model Context Protocol (MCP), Google ADK, ReAct Pattern, Reliable Agent Workflows, State Graph Architectures, Task Decomposition, Conditional Routing, Parallel Fan-out/Gather, Agent Memory & Context Management, Fallback & Graceful Degradation

Evaluation & Reliability: AI Evaluation Frameworks, Agent Trajectory Evaluation, Benchmarking, Experiment Tracking (MLflow, Weights & Biases), Output Validation, Hallucination Mitigation, Source Grounding, Confidence Scoring, Observability & Monitoring, Deterministic Test Suites (pytest), Bounded Execution & Retry Discipline

Languages & Backend: Python, TypeScript, JavaScript, Java, SQL, Bash, FastAPI, REST API Design, Microservices, Async Processing, Model-Serving APIs, PostgreSQL, Supabase, Redis, Git/GitHub, Linux

LLMs & Retrieval: OpenAI GPT-4o, Azure OpenAI, LLaMA 3.3-70B (Groq), GPT4All (self-hosted), Prompt & Context Engineering, Retrieval-Augmented Generation (RAG), Embeddings, Semantic & Vector Search (Qdrant, FAISS), Hybrid BM25 + Vector Retrieval, Cross-Encoder Re-Ranking, Query Routing, Sentence Transformers

Deployment & Tooling: Docker, Containerized Cloud Deployment, CI/CD (GitHub Actions, CircleCI), Microsoft Azure, Model Serving, AI-Assisted Development (Claude Code, Codex), React / Next.js, Streamlit, Gradio, Code Review, Technical Documentation

EXPERIENCE

AI Engineer — NBCUniversal *Jan 2025 – Present*

- Built and deployed production AI agents and copilots in Python with LangGraph and LangChain — orchestrating planning, retrieval, tool-execution, and validation agents into reliable multi-step workflows that automate enterprise processes.
- Developed tool orchestration and integrations connecting agents to external APIs, structured data, and enterprise systems, returning structured, validatable outputs at each step.
- Instrumented agent workflows with AI evaluation frameworks, monitoring, and CI/CD, so reliability and answer quality were measured rather than assumed.

Research Assistant — Texas Tech University *Aug 2024 – May 2026*

- Built reproducible AI/ML and LLM pipelines in Python (PyTorch, TensorFlow, Scikit-learn) spanning data preprocessing, feature engineering, model training, and evaluation for research-driven analytics.
- Prototyped and benchmarked NLP, deep-learning, and Transformer/LLM models — running inference evaluation, interpretability, and result validation to compare approaches objectively.
- Automated experiment tracking, model evaluation, and technical documentation to improve reproducibility and MLOps readiness across experiments.

Programmer Analyst Trainee — Cognizant *Jun 2022 – Jun 2024*

- Shipped customer-facing LLM features into enterprise workflows, translating user problems into production-grade AI solutions.
- Built RAG pipelines over structured and unstructured enterprise data with Azure Cognitive Search and the REST services behind them, grounding outputs so answers met accuracy expectations.
- Tested and debugged LLM inference and retrieval components end to end, improving response reliability against real user tasks.

PROJECTS

Multi-Agent Travel Copilot Case Study

Python, LangGraph, LangChain, FastAPI, Streamlit, OpenAI

- Architected a multi-agent system in LangGraph — 11 agents coordinated through a shared AtlasState TypedDict, an explicit memory contract with namespaced keys that lets agents decompose a goal and hand off context without collisions or drift.
- Integrated 6 external APIs as agent tools with structured, validatable outputs, orchestrating multi-step calls with parallel fan-out/gather execution.
- Shipped conditional routing and graph-level error recovery so multi-step workflows degrade predictably rather than failing outright when a tool call errors or times out.

AI Research Assistant — RAG Document Intelligence Live Demo

Python, FastAPI, Qdrant, Sentence Transformers, LLaMA 3.3-70B (Groq), Redis, Streamlit

- Deployed a 3-tier service (Streamlit → FastAPI on Render → Qdrant Cloud) with a Redis caching layer between inference and retrieval, using hybrid BM25 + vector search and cross-encoder re-ranking to boost answer accuracy.
- Mitigated hallucination directly: enforced NOT_FOUND grounding with three-level fallback and heuristic confidence scoring, so the system declines rather than fabricates.
- Designed a query router that selects between web and RAG retrieval per request, controlling what enters the context window and keeping inference cost bounded.

AI Portfolio Forecasting & Optimization Pipeline Live Demo

Python, Prophet, SciPy, Supabase, GitHub Actions, CircleCI

- Built and deployed a scheduled production pipeline on GitHub Actions — Prophet forecasting plus SciPy SLSQP optimization — persisted to Postgres and surfaced on a monitored live dashboard.
- Enforced CI quality gates via CircleCI parallel stages (lint, type check, pytest) so every scheduled run is reproducible.
- Forecast prices with Prophet time-series models and optimized a 12-ticker portfolio under constraints via SciPy SLSQP, with the daily run fully automated end to end.

EDUCATION

M.S., Computer Science — Texas Tech University *May 2026*

Coursework: Machine Learning · Neural Networks · Advanced Data Science · Algorithms · Software Modeling & Architecture